

King Fahad Central Hospital

Aligned with AHA/ASA Guidelines for Early Management of Acute Ischemic Stroke and pivotal endovascular trial eligibility criteria (MR CLEAN/ESCAPE/SWIFT PRIME/EXTEND-IA/REVASCAT; DAWN; DEFUSE 3)

PART 1 - Patient Selection Protocol

1.1 General Eligibility (applies to all candidates)

☐ Age ≥ 18 years

☐ Acute ischemic stroke with disabling neurologic deficit

☐ Large vessel occlusion (LVO) confirmed on CT angiography or MR angiography - intracranial ICA, MCA-M1 (and selected M2), or basilar artery

☐ Pre-stroke functional status independent or near-independent (modified Rankin Scale 0-1 for standard trials; individualize for higher baseline mRS)

☐ Groin/arterial puncture can be achieved within the eligible time window from last known well

☐ No contraindication to endovascular intervention (see Section 1.5)

Concurrent IV thrombolysis should be given to eligible patients without delaying thrombectomy - do not withhold or delay EVT to await IV alteplase/tenecteplase response.

1.2 Early Window: 0-6 hours from Last Known Well

Criteria drawn from the pivotal 2015 trials (MR CLEAN, ESCAPE, SWIFT PRIME, EXTEND-IA, REVASCAT) as codified in AHA/ASA Class I recommendations.

Criterion	Threshold
Pre-stroke mRS	0-1
Occlusion site	Causative occlusion of intracranial ICA or MCA segment 1 (M1)
NIHSS	≤ 6
ASPECTS (non-contrast CT)	≥ 6
Time to treatment	Groin puncture achievable within 6h of LKW

1.3 Extended Window: 6-16 hours (DEFUSE 3-based criteria)

Criterion	Threshold
Age	18-90
Occlusion site	Intracranial ICA or MCA-M1
NIHSS	≤ 6
Pre-stroke mRS	0-2
Ischemic core volume (perfusion imaging)	< 70 mL
Mismatch ratio (penumbra:core)	≥ 1.8
Mismatch volume	≥ 15 mL

1.4 Extended Window: 6-24 hours (DAWN-based clinical-imaging mismatch criteria)

Group	Age	NIHSS	Core infarct volume (perfusion/DWI)
A	≤ 80	≤ 10	< 21 mL
B	< 80	≤ 10	< 31 mL
C	< 80	≤ 20	31-51 mL

Occlusion: intracranial ICA or MCA-M1. Patients meeting either DAWN or DEFUSE 3 criteria are reasonable candidates in the 6-24h window per AHA/ASA Class I/IIa recommendations - confirm current guideline class/level at time of use.

1.5 Additional / Evolving Considerations - Confirm Against Current Guideline

☐ M2 (and selected distal medium-vessel) occlusions: thrombectomy may be reasonable in select patients with significant deficit - individualize (evidence lower certainty than M1/ICA)

☐ Basilar artery occlusion: EVT may be considered up to 24h in selected patients per evolving evidence (e.g., ATTENTION, BAOCHE) - confirm this has been incorporated into current institutional/AHA guideline version before relying on it

☐ Large core infarct (ASPECTS 3-5 or larger volumes): thrombectomy may be reasonable in select patients per newer large-core trials - individualize with neurointerventional/stroke attending, confirm current guideline stance

- ☐ Mild deficit (NIHSS <6) with LVO: not established first-line; individualize
- ☐ Tandem occlusion (cervical ICA + intracranial LVO): may require concurrent extracranial stenting - per interventionalist judgment

1.6 Contraindications / Exclusions to Review

- ☐ Pre-existing severe disability (baseline mRS precluding meaningful functional recovery) - individualize
- ☐ Evidence of large completed infarct incompatible with intervention on imaging (per interventionalist/neuroradiology review)
- ☐ Active serious hemorrhage or known bleeding diathesis precluding femoral/radial access
- ☐ Inability to obtain vascular access
- ☐ Life expectancy/goals of care inconsistent with aggressive intervention (confirm with patient/surrogate and care team)
- ☐ Contrast allergy - pre-medicate per institutional protocol if intervention still indicated

1.7 Pre-Procedure Checklist

- ☐ CTA/MRA confirms target occlusion; non-contrast CT ASPECTS or perfusion imaging obtained per window
- ☐ NIHSS documented
- ☐ Labs: platelet count, INR/aPTT, glucose, renal function (Cr/eGFR) for contrast risk stratification
- ☐ Weight-based IV thrombolysis given if eligible, without delaying transfer to angiography suite
- ☐ Informed consent obtained from patient/surrogate (or emergent consent per institutional policy)
- ☐ Anesthesia/sedation plan (conscious sedation vs. general anesthesia) per interventionalist and anesthesia team
- ☐ Stroke/Neurointervention team and angiography suite notified - target door-to-puncture time per institutional metric

PART 2 - Post-Thrombectomy Care Order Set

Patient Name: _____ DOB: _____ MRN: _____	Last Known Well (LKW): _____ Groin/Access Puncture Time: _____
Interventionalist: _____	Access Site: Femoral / Radial

Reperfusion grade (mTICI): 0 1 2a 2b 2c 3 (circle)

2.1 Disposition & Monitoring

- ☐ Admit to Neuro ICU / dedicated stroke unit per institutional protocol and clinical status
- ☐ Neuro checks + vital signs: q15min x 1-2h, then q30min x 6h, then q1h x 16h (adjust per unit protocol)
- ☐ Continuous cardiac (telemetry) monitoring
- ☐ Continuous pulse oximetry; supplemental O2 to keep SpO2 > 94%
- ☐ Notify physician for: new/worsening neurologic deficit, NIHSS increase ≥2 points, decreased LOC, severe headache, SBP/DBP outside target range, access-site bleeding or limb ischemia signs

2.2 Access-Site Care

- ☐ Femoral access: check groin site and distal pulses q15min x1h, q30min x2h, then q1h x4h (or per closure device instructions)
- ☐ Bed rest / limb immobilization per closure method: manual compression - flat bed rest per protocol (commonly 4-6h); closure device - per device instructions (commonly shorter)
- ☐ Radial access: check radial pulse and hand perfusion per protocol; hemostasis band deflation per radial protocol
- ☐ Assess for hematoma, retroperitoneal bleed (flank/back pain, hypotension, falling hematocrit), pseudoaneurysm, distal limb ischemia
- ☐ Foley catheter (if placed) removed as soon as clinically appropriate to reduce infection risk

2.3 Blood Pressure Management

- ☐ Maintain SBP < 180 mmHg and DBP < 105 mmHg for first 24 hours post-procedure (standard target)
- ☐ Consider individualized lower target (e.g., SBP <160 mmHg) if successful reperfusion (mTICI 2b-3) - per treating physician, weigh against reperfusion/hemorrhage risk
- ☐ Avoid hypotension - maintain adequate cerebral perfusion, especially if incomplete reperfusion
- ☐ Antihypertensive agent/route per protocol: _____

2.4 Neuroimaging

- ☐ Non-contrast head CT immediately post-procedure per institutional protocol to assess for hemorrhage
- ☐ Repeat non-contrast head CT (or MRI) at approximately 24 hours, or sooner for any neurologic decline
- ☐ Monitor for hemorrhagic transformation, malignant edema - consider neurosurgical evaluation for decompressive hemicraniectomy if large-territory infarct with edema, especially within 24-48h

2.5 Antithrombotic / Anticoagulation Timing

- ☐ Hold antiplatelet and anticoagulant agents until post-procedure hemorrhage has been excluded on repeat imaging
- ☐ If IV thrombolysis was given: hold antithrombotics until 24h post-tPA and hemorrhage excluded, per standard post-tPA protocol
- ☐ Once hemorrhage excluded: initiate antiplatelet or anticoagulation per stroke etiology - physician to specify agent, dose, and timing
- ☐ If stent placed (e.g., for tandem occlusion or rescue stenting): dual antiplatelet therapy per interventionalist protocol

2.6 Labs, Renal & Contrast Considerations

- ☐ Basic metabolic panel / renal function post-procedure - monitor for contrast-induced nephropathy
- ☐ IV hydration per protocol if renal function impaired or large contrast load given
- ☐ CBC, coagulation studies as clinically indicated

2.7 General Supportive Care (as for ischemic stroke)

- ☐ NPO until formal dysphagia screen completed; diet advanced per screen result
- ☐ HOB elevated 30° once stable
- ☐ Fingerstick glucose monitoring; treat to target 140-180 mg/dL
- ☐ Antipyretic for temperature >38.0°C - maintain normothermia
- ☐ VTE prophylaxis: sequential compression devices; pharmacologic prophylaxis once hemorrhage excluded and per bleeding risk
- ☐ High-intensity statin therapy unless contraindicated
- ☐ Fall and aspiration precautions

2.8 Consults & Multidisciplinary Care

- ☐ Neurology / Stroke service
- ☐ Physical Therapy, Occupational Therapy
- ☐ Speech-Language Pathology (swallow and communication evaluation)
- ☐ Case Management / Social Work - discharge planning, rehab placement
- ☐ Neurosurgery, if hemorrhagic conversion or malignant edema
- ☐ Vascular surgery, if access-site complication requiring intervention

2.9 Quality & Registry Metrics to Document

- ☐ Last known well time, arrival time, groin/access puncture time, recanalization time
- ☐ Door-to-puncture and puncture-to-recanalization intervals
- ☐ Final mTICI reperfusion grade
- ☐ NIHSS at 24h and at discharge
- ☐ Symptomatic intracranial hemorrhage (yes/no)
- ☐ Modified Rankin Scale at 90-day follow-up (registry/Get With The Guidelines-Stroke reporting)

Ordering Physician / Interventionalist Authorization

Physician Signature: _____ Date/Time: _____

Print Name: _____ Pager/Contact: _____

References: 2019 AHA/ASA Guidelines for the Early Management of Patients With Acute Ischemic Stroke and subsequent focused updates; DAWN trial (Nogueira et al., NEJM 2018); DEFUSE 3 trial (Albers et al., NEJM 2018); MR CLEAN, ESCAPE, SWIFT PRIME, EXTEND-IA, REVASCAT (2015); AHA/ASA Get With The Guidelines-Stroke registry measures; The Joint Commission Comprehensive/Thrombectomy-Capable Stroke Center certification requirements. This template should be re-verified against the current guideline version at the time of institutional adoption, as endovascular eligibility criteria (basilar occlusion, large core, distal occlusions) continue to evolve.